Q1.

The waiting times for patients to see a doctor in a hospital can be modelled with a normal distribution with known variance of 10 minutes.

(a) A random sample of 100 patients has a total waiting time of 3540 minutes.

Calculate a 98% confidence interval for the population mean of waiting times, giving values to four significant figures.

(4)

(b) Dante conducts a hypothesis test with the sample from part **(a)** on the waiting times. Dante's hypotheses are

$$H_0 : \mu = 38$$
$$H_1 : \mu \neq 38$$

Dante uses a 2% level of significance.

Explain whether Dante accepts or rejects the null hypothesis.

(1)

(Total 5 marks)

Q2.

David, a zoologist, is investigating a particular species of monitor lizard. He measures the lengths, in centimetres, of a random sample of this particular species of lizard. His measured lengths are

53.2 57.8 55.3 58.9 59.0 60.2 61.8 62.3 65.4 66.5

The lengths may be assumed to be normally distributed.

David correctly constructed a 90% confidence interval for the mean length of lizard using the measured lengths given and the formula $\bar{x} \pm \left(b \times \frac{s}{\sqrt{n}}\right)$

This interval had limits of 57.63 and 62.45, correct to two decimal places.

(a) State the value for b used in David's formula.

(1)

(b) David interprets his interval and states,

"My confidence interval indicates that exactly 90% of the population of lizard lengths for this particular species lies between 57.63 cm and 62.45 cm".

Do you think David's statement is true? Explain your reasoning.

(2)

(c) David's assistant, Amina, correctly constructs a β% confidence interval from David's random sample of measured lengths.

Amina informs David that the width of her confidence interval is 8.54.

Find the value of β.

(3)
(Total 6 marks)

Q3.

Dimitra is an athlete who competes in 400 m races. The times, in seconds, for her first six races of the 2012 season were

54.86 53.09 53.75 52.88 51.97 51.81

(a) Assuming that these data form a random sample from a normal distribution, construct a 95% confidence interval for the mean time of Dimitra's races in the 2012 season, giving the limits to two decimal places.

(5)

(b) For the 2011 season, Dimitra's mean time for her races was 53.41 seconds. After her first six races of the 2012 season, her coach claimed that the data showed that she would be more successful in races during the 2012 season than during the 2011 season. Make **two** comments about the coach's claim.

(2)
(Total 7 marks)

Q4.

The weight of sand in a bag can be modelled by a normal random variable with unknown mean μ kilograms and known standard deviation 0.4 kilograms.

The sand in each of a random sample of 25 bags from a large batch is weighed. The mean weight of sand in these 25 bags is found to be 19.9 kg.

(a) Construct a 98% confidence interval for the mean weight of sand in bags in the batch.

(4)

(b) Hence comment on the claim that bags in the batch contain an average of 20 kg of sand.

(2)
(Total 6 marks)

Q5.

The weight, X kilograms, of sand in a bag can be modelled by a normal random variable with unknown mean μ and known standard deviation 0.4.

(a) The sand in each of a random sample of 25 bags from a large batch is weighed. The **total** weight of sand in these 25 bags is found to be 497.5 kg.

(i) Construct a 98% confidence interval for the mean weight of sand in bags in the batch.

(5)

(ii) Hence comment on the claim that bags in the batch contain an average of 20 kg of sand.

(2)

(b) The weight, Y kilograms, of cement in a bag can be modelled by a normal random variable with mean 25.25 and standard deviation 0.35.

A firm purchases 10 such bags. These bags may be considered to be a random sample.

(i) Determine the probability that the **mean** weight of cement in the 10 bags is **less** than 25 kg.

(4)

(ii) Calculate the probability that the weight of cement in **each** of the 10 bags is **more** than 25 kg.

(4)

(Total 15 marks)

Q6.

Gemma, a biologist, studies guillemots, which are a species of seabird. She has found that the weight of an adult guillemot may be modelled by a normal distribution with mean μ grams. During 2012, she measured the weight, x grams, of each of a random sample of 9 adult guillemots and obtained the following results.

$$\sum x = 8532 \qquad \text{and} \qquad \sum (x - \bar{x})^2 = 38\,538$$

(a) Construct a 98% confidence interval for μ based on these data.

(5)

(b) The corresponding confidence interval for μ obtained by Gemma based on a random sample of 9 adult guillemots measured during 2011 was (927, 1063), correct to the nearest gram.

(i) Find the mean weight of guillemots in this sample.

(1)

(ii) Studies of some other species of seabird have suggested that their mean weights were less in 2012 than in 2011. Comment on whether Gemma's two confidence intervals provide evidence that the mean weight of guillemots was less in 2012 than in 2011.

(2)

(Total 8 marks)

Q7.

A random sample of 50 full-time university employees was selected as part of a higher education salary survey.

The annual salary in thousands of pounds, x, of each employee was recorded, with the following summarised results.

$$\bar{x} = 45.8 \qquad \text{and} \qquad s = 24.0$$

Also recorded was the fact that 6 of the 50 salaries exceeded £60 000.

(a) Show why the annual salary, X, of a full-time university employee is unlikely to be normally distributed. Give numerical support for your answer.

(2)

(b) (i) Indicate why the mean annual salary, $\overline{X}$, of a random sample of 50 full-time university employees may be assumed to be normally distributed.

(2)

(ii) Hence construct a 99% confidence interval for the mean annual salary of full-time university employees.

(4)

(c) It is claimed that the annual salaries of full-time university employees have an average which exceeds £55 000 and that more than 25% of such salaries exceed £60 000.

Comment on **each** of these two claims.

(3)

(Total 11 marks)

Q8.

A random sample of 50 full-time university employees was selected as part of a higher education salary survey.

The annual salary in thousands of pounds, x, of each employee was recorded, with the following summarised results.

$$\sum x = 2290.0 \quad \text{and} \quad \sum (x - \overline{x})^2 = 28\,225.50$$

Also recorded was the fact that 6 of the 50 salaries exceeded £60 000.

(a) (i) Calculate values for $\overline{x}$ and s, where s^2 denotes the unbiased estimate of σ^2.

(3)

(ii) Hence show why the annual salary, X, of a full-time university employee is unlikely to be normally distributed. Give numerical support for your answer.

(2)

(b) Assuming that the mean annual salary, $\overline{X}$, of a random sample of 50 full-time university employees is normally distributed, construct a 99% confidence interval for the mean annual salary of full-time university employees.

(4)

(c) It is claimed that the annual salaries of full-time university employees have an average which exceeds £55 000 and that more than 25% of such salaries exceed £60 000.

Comment on **each** of these two claims.

(3)

(Total 12 marks)

Q9.

The volume of bleach in a 5-litre bottle may be modelled by a random variable with a standard deviation of 75 millilitres.

The volume, in litres, of bleach in each of a random sample of 36 such bottles was measured. The 36 measurements resulted in a mean volume of 5.05 litres and exactly 8 bottles contained less than 5 litres.

(a) Construct a 98% confidence interval for the mean volume of bleach in a 5-litre bottle.

(4)

(b) It is claimed that the mean volume of bleach in a 5-litre bottle exceeds 5 litres and also that fewer than 10 per cent of such bottles contain less than 5 litres.

Comment, with numerical justification, on **each** of these two claims.

(3)
(Total 7 marks)

Q10.

At the start of the 2012 season, the ages of the members of the Warwickshire Acorns Cricket Club could be modelled by a normal random variable, X years, with mean μ and standard deviation σ.

The ages, x years, of a random sample of 15 such members are summarised below.

$$\sum x = 546 \quad \text{and} \quad \sum (x - \bar{x})^2 = 1407.6$$

(a) Construct a 98% confidence interval for μ, giving the limits to one decimal place.

(6)

(b) At the start of the 2005 season, the mean age of the members was 40.0 years.

Use your confidence interval constructed in part (a) to indicate, with a reason, whether the mean age had changed.

(2)
(Total 8 marks)

Q11.

Becky owns a taxi. Each weekday morning, she collects Steve from his home and takes him to the train station.

A record of the times, x minutes, for a random sample of 65 such taxi journeys is summarised by

$$\sum x = 1326.0 \quad \text{and} \quad \sum (x - \bar{x})^2 = 400.24$$

(a) (i) Calculate the value of the sample mean, x.

(1)

(ii) Show that, correct to two decimal places, s = 2:50, where s^2 denotes the unbiased estimate of the population variance.

(2)

(b) Construct a 96% confidence interval for the mean journey time.

(4)

(c) Comment on Becky's claim that the mean journey time is more than 20 minutes.

(2)
(Total 9 marks)

Q12.

The volume, X litres, of orange juice in a 1-litre carton may be modelled by a normal distribution with unknown mean m.

The volumes, x litres, recorded to the nearest 0.01 litre, in a random sample of 100 cartons are shown in the table.

Volume (x litres)	**Number of cartons (f)**
0.95 – 0.97	2
0.98 – 1.00	7
1.01 – 1.03	15
1.04 – 1.06	32
1.07 – 1.09	22
1.10 – 1.12	14
1.13 – 1.15	7
1.16 – 1.18	1
Total	**100**

(a) For the group '0.98 – 1.00':

(i) show that it has a mid-point of 0.99 litres;

(ii) state the minimum and the maximum values of x that could be included in this group.

(2)

(b) Calculate, to three decimal places, estimates of the mean and the standard deviation of these 100 volumes.

(3)

(c) (i) Construct an approximate 99% confidence interval for m.

(4)

(ii) Give a reason why the confidence interval is approximate rather than exact.

(1)

(d) Give a reason in support of the claim that:

(i) $\mu > 1$;

(ii) P(0.94 < X < 1.16) is approximately 1.

(2)
(Total 12 marks)

Q13.

A factory produces bottles of brown sauce and bottles of tomato sauce.

(a) The content, Y grams, of a bottle of brown sauce is normally distributed with mean μ_Y and variance 4.

A quality control inspection found that the mean content, y grams, of a random sample of 16 bottles of brown sauce was 450.

Construct a 95% confidence interval for μ_Y.

(3)

(b) The content, X grams, of a bottle of tomato sauce is normally distributed with mean μ_X and variance σ^2.

A quality control inspection found that the content, x grams, of a random sample of 9 bottles of tomato sauce was summarised by

$$\sum x = 4950 \quad \text{and} \quad \sum (x - \bar{x})^2 = 334$$

(i) Construct a 90% confidence interval for μ_X.

(5)

(ii) Holly, the supervisor at the factory, claims that the mean content of a bottle of tomato sauce is 545 grams.

Comment, with a justification, on Holly's claim. State the level of significance on which your conclusion is based.

(3)

(Total 11 marks)

Q14.

Rice that can be cooked in microwave ovens is sold in packets which the manufacturer claims contain a mean weight of more than 250 grams of rice.

The weight of rice in a packet may be modelled by a normal distribution.

A consumer organisation's researcher weighed the contents, x grams, of each of a random sample of 50 packets. Her summarised results are:

$$\bar{x} = 251.1 \quad \text{and} \quad s = 1.94$$

(a) (i) Construct a 96% confidence interval for the mean weight of rice in a packet, giving the limits to one decimal place.

(4)

(ii) Hence comment on the manufacturer's claim.

(2)

(b) (i) Construct an interval within which approximately 96% of the weights of rice in individual packets will lie.

(2)

(ii) The statement ‘250 grams’ is printed on each packet.

Comment on what your interval in part (b)(i) reveals about this statement.

(1)

(Total 9 marks)

Q15.

Rice that can be cooked in microwave ovens is sold in packets which the manufacturer claims contain a mean weight of more than 250 grams of rice.

The weight of rice in a packet may be modelled by a normal distribution.

A consumer organisation’s researcher weighed the contents, x grams, of each of a random sample of 50 packets. Her summarised results are:

$$\bar{x} = 251.1 \quad \text{and} \quad \sum (x - \bar{x})^2 = 184.5$$

(a) Show that, correct to two decimal places, s = 1:94, where s^2 denotes the unbiased estimate of the population variance.

(1)

(b) (i) Construct a 96% confidence interval for the mean weight of rice in a packet, giving the limits to one decimal place.

(4)

(ii) Hence comment on the manufacturer’s claim.

(2)

(c) The statement ‘250 grams’ is printed on each packet.

Explain, with reference to the values of $\bar{x}$ and s, why the consumer organisation may consider this statement to be dubious.

(2)

(Total 9 marks)

Q16.

Cherie Glass is a salesperson for RDW Ltd, a replacement door and window company. Her supervisor, Wyn Doe, believes that Cherie spends too long discussing options with potential clients. With this in mind, he records the times, t minutes, that Cherie spends in discussions with 50 potential clients.

From these 50 times, Wyn finds that

$$\sum t = 3155 \qquad \sum (t - \bar{t})^2 = 7180.5$$

(a) Stating a necessary assumption about the 50 times, calculate unbiased estimates of the mean, μ, and the variance, σ^2, for the time, in minutes, that Cherie spends with potential clients. Give your answers to one decimal place.

(3)

(b) Hence construct a 99% confidence interval for μ.

(4)

(c) Given that RDW Ltd expects its salespeople to spend, on average, no longer than 60 minutes with potential clients, comment on Wyn's belief.

(2)

(d) Given that μ = 62, state the probability that a 99% confidence interval for μ will **not** contain 62.

(1)
(Total 10 marks)

Q17.

The weight of gravel, in kilograms, collected by a power shovel may be modelled by a normal distribution with unknown mean, μ, and standard deviation 50.

The weights of a random sample of 20 collections have a mean of 1030 kg.

(a) Construct a 98% confidence interval for μ, giving the limits to four significant figures.

(4)

(b) Comment on a claim that the power shovel is, on average, collecting more than 1000 kg of gravel.

(2)
(Total 6 marks)

Q18.

The height, in metres, of adult male African elephants may be assumed to be normally distributed with mean μ and standard deviation 0.20.

The heights of a sample of 12 such elephants were measured with the following results, in metres.

3.37 3.45 2.93 3.42 3.49 3.67 2.96 3.57 3.36 2.89 3.22 2.91

(a) Stating a necessary assumption, construct a 98% confidence interval for μ.

(6)

(b) The mean height of adult male **Asian** elephants is known to be 2.90 metres.

Using your confidence interval, state, with a reason, what can be concluded about the mean heights of adult males in these two types of elephant.

(2)
(Total 8 marks)

Q19.

A speed camera was used to measure the speed, V mph, of John's serves during a tennis singles championship.

For 10 randomly selected serves,

$$\sum v = 1179 \quad \text{and} \quad \sum (v - \bar{v})^2 = 1014.9$$

where $\bar{v}$ is the sample mean.

(a) Construct a 99% confidence interval for the mean speed of John's serves at this tennis championship, stating any assumption that you make.

(7)

(b) Hence comment on John's claim that, at this championship, he consistently served at speeds in excess of 130 mph.

(1)
(Total 8 marks)

Q20.

Vernon, a service engineer, is expected to carry out a boiler service in one hour.

One hour is subtracted from each of his actual times, and the resulting differences, x minutes, for a random sample of 100 boiler services have a mean, $\bar{x}$, of 1.90 and a standard deviation, s, of 3.32 .

(a) Deduce, in minutes, the mean and the standard deviation of Vernon's actual service times for this sample.

(3)

(b) Construct a 98% confidence interval for the mean time taken by Vernon to carry out a boiler service.

(4)

(c) Vernon claims that, on average, a boiler service takes much longer than an hour.

Comment, with a justification, on this claim.

(1)
(Total 8 marks)

Q21.

Vernon, a service engineer, is expected to carry out a boiler service in one hour.

One hour is subtracted from each of his actual times, and the resulting differences, x minutes, for a random sample of 100 boiler services are summarised in the table.

Difference	Frequency
$-6 \le x < -4$	4
$-4 \le x < -2$	9
$-2 \le x < 0$	13
$0 \le x < 2$	27
$2 \le x < 4$	21

$4 \le x < 6$	15
$6 \le x < 8$	7
$8 \le x \le 10$	4
Total	**100**

(a) (i) Calculate estimates of the mean and the standard deviation of these differences.

(4)

(ii) Hence deduce, in minutes, estimates of the mean and the standard deviation of Vernon's actual service times for this sample.

(3)

(b) (i) Construct an approximate 98% confidence interval for the mean time taken by Vernon to carry out a boiler service.

(4)

(ii) Give a reason why this confidence interval is approximate rather than exact.

(1)

(c) Vernon claims that, more often than not, a boiler service takes more than an hour and that, on average, a boiler service takes much longer than an hour.

Comment, with a justification, on **each** of these claims.

(2)

(Total 14 marks)

Q22.

The weight of fat in a digestive biscuit is known to be normally distributed.

Pat conducted an experiment in which she measured the weight of fat, x grams, in each of a random sample of 10 digestive biscuits, with the following results:

$$\sum x = 31.9 \quad \text{and} \quad \sum (x - \bar{x})^2 = 1.849$$

(a) (i) Construct a 99% confidence interval for the mean weight of fat in digestive biscuits.

(5)

(ii) Comment on a claim that the mean weight of fat in digestive biscuits is 3.5 grams.

(2)

(b) If 200 such 99% confidence intervals were constructed, how many would you expect **not** to contain the population mean?

(1)

(Total 8 marks)

Q23.

A very popular play has been performed at a London theatre on each of 6 evenings per week for about a year. Over the past 13 weeks (78 performances), records have been kept of the proceeds from the sales of programmes at each performance. An analysis of these records has found that the mean was £184 and the standard deviation was £32.

(a) Assuming that the 78 performances may be considered to be a random sample, construct a 90% confidence interval for the mean proceeds from the sales of programmes at an evening performance of this play.

(4)

(b) Comment on the likely validity of the assumption in part (a) when constructing a confidence interval for the mean proceeds from the sales of programmes at an evening performance of:

(i) this particular play;

(ii) any play.

(3)
(Total 7 marks)

Q24.

(a) The length, X centimetres, of adult male eels in a river may be assumed to be normally distributed with a mean of 38 and a standard deviation of 5.

Determine:

(i) $P(X < 40)$;

(2)

(ii) $P(30 < X < 40)$;

(3)

(iii) the length exceeded by 75% of adult male eels in the river.

(4)

(b) A sample of 40 adult female eels was taken at random from the river and the length of each eel was measured.

The mean and standard deviation of these lengths were found to be 107 cm and 19.1 cm respectively.

(i) Construct a 98% confidence interval for the mean length of adult female eels in the river.

(4)

(ii) Hence comment on a claim that, in this river, the average length of adult female eels is more than $2\frac{1}{2}$ times that of adult male eels.

(3)
(Total 16 marks)

Q25.

(a) A sample of 50 washed baking potatoes was selected at random from a large batch. The weights of the 50 potatoes were found to have a mean of 234 grams and a standard deviation of 25.1 grams.

Construct a 95% confidence interval for the mean weight of potatoes in the batch.

(4)

(b) The batch of potatoes is purchased by a market stallholder. He sells them to his customers by allowing them to choose any 5 potatoes for £1.

Give a reason why such chosen potatoes are unlikely to represent a random sample from the batch.

(1)
(Total 5 marks)

Q26.

Members of a residents' association are concerned about the speeds of cars travelling through their village. They decide to record the speed, in mph, of each of a random sample of 10 cars travelling through their village, with the following results:

33 27 34 30 48 35 34 33 43 39

(a) Construct a 99% confidence interval for μ, the mean speed of cars travelling through the village, stating any assumption that you make.

(7)

(b) Comment on the claim that a 30 mph speed limit is being adhered to by most motorists.

(3)
(Total 10 marks)

Q27.

When an alarm is raised at a market town's fire station, the fire engine cannot leave until at least five fire-fighters arrive at the station. The call-out time, X minutes, is the time between an alarm being raised and the fire engine leaving the station.

The value of X was recorded on a random sample of 50 occasions. The results are summarised below, where $\bar{x}$ denotes the sample mean.

$$\sum x = 286.5 \qquad \sum (x - \bar{x})^2 = 45.16$$

(a) Find values for the mean and standard deviation of this sample of 50 call-out times.

(2)

(b) Hence construct a 99% confidence interval for the mean call-out time.

(4)

(c) The fire and rescue service claims that the station's mean call-out time is less than 5 minutes, whereas a parish councillor suggests that it is more than $6\frac{1}{2}$ minutes.

Comment on **each** of these claims.

(2)
(Total 8 marks)

Q28.

The weight, W grams, of rice contained in a packet can be modelled by a normal distribution with mean μ and variance σ^2.

The weights, in grams, of rice contained in each of a random sample of 10 packets were

301.2	305.6	298.4	297.7	292.6
306.1	296.8	299.2	304.0	308.4

(a) Calculate unbiased estimates for μ and σ^2.

(2)

(b) Hence construct a 99% confidence interval for μ, giving the limits to one decimal place.

(5)

(c) The packets are claimed to contain, on average, at least 290 grams of rice.

Comment on this claim.

(2)
(Total 9 marks)

Q29.

The time, T minutes, that parents have to wait before seeing a mathematics teacher at a school parents' evening can be modelled by a normal distribution with mean μ and standard deviation σ.

At a recent parents' evening, a random sample of 9 parents was asked to record the times that they waited before seeing a mathematics teacher.

The times, in minutes, are

5 12 10 8 7 6 9 7 8

(a) Construct a 90% confidence interval for μ.

(7)

(b) Comment on the headteacher's claim that the mean time that parents have to wait before seeing a mathematics teacher is 5 minutes.

(2)
(Total 9 marks)

Q30.

Currants are sold in 1000-gram packets and in 500-gram packets.

(a) The weights of 1000-gram packets may be assumed to be normally distributed with a mean of 1012 grams and a standard deviation of 5 grams.

Determine the probability that a randomly selected packet weighs:

(i) less than 1015 grams;

(3)

(ii) more than 1005 grams;

(3)

(iii) between 1005 grams and 1015 grams.

(2)

(b) The weight, y grams, of each of a random sample of fifty 500-gram packets of currants was recorded with the following results, where $\bar{y}$ denotes the sample mean:

$n = 50$ $\sum y = 25\,142.5$ $\sum (y - \bar{y})^2 = 2519.0361$

Number of packets weighing less than 500 grams = 6

(i) Construct a 98% confidence interval for the mean weight of 500-gram packets of currants, giving the limits to two decimal places.

(6)

(ii) On each packet it states 'Contents 500 grams'.

Comment on this statement using **both** the given information **and** your confidence interval.

(3)

(Total 17 marks)

Q31.

The weights of packets of sultanas may be assumed to be normally distributed with a standard deviation of 6 grams.

The weights of a random sample of 10 packets were as follows:

498	496	499	511	503	505	510	509	513	508

(a) (i) Construct a 99% confidence interval for the mean weight of packets of sultanas, giving the limits to one decimal place.

(5)

(ii) On each packet it states 'Contents 500 grams'.

Comment on this statement using **both** the given sample **and** your confidence interval.

(3)

(b) Given that the mean weight of all packets of sultanas is 500 grams, state the probability that a 99% confidence interval for the mean, calculated from a random sample of packets, will **not** contain 500 grams.

(1)
(Total 9 marks)

Q32.

The volume, in millilitres, of lemonade in mini-cans may be assumed to be normally distributed with a standard deviation of 3.5.

The volumes, in millilitres, of lemonade in a random sample of 12 mini-cans were as follows.

155	148	156	149	147	156
157	156	150	154	148	154

(a) Construct a 98% confidence interval for the mean volume of lemonade in a mini-can, giving the limits to one decimal place.

(5)

(b) On each mini-can is printed "150 ml". Comment on this, using the given sample and your confidence interval in part (a).

(3)
(Total 8 marks)

Q33.

Each morning, Mustafa either cycles or walks to school.

(a) The time, X minutes, taken by Mustafa to cycle to school maybe modelled by a normal random variable with mean 12 and standard deviation 2.5.

Determine:

(i) $P(X < 15)$;

(2)

(ii) $P(10 < X < 15)$.

(3)

(b) On each of a random sample of 50 mornings, Mustafa records his time, y minutes, to walk to school. From his recordings, Mustafa calculates the following quantities, where $\bar{y}$ denotes the sample mean.

$$\sum y = 835.0 \qquad \sum (y - \bar{y})^2 = 533.61$$

(i) Construct a 99% confidence interval for the mean time taken by Mustafa to walk to school.

(6)

(ii) Mustafa suspects that, on average, the time he takes to walk to school is more than 25% longer than the time he takes to cycle to school.

State, with reasons, whether this suspicion is supported by your confidence interval.

(3)

(Total 14 marks)

Q34.

The weights of oranges sold by a greengrocer are known to be normally distributed. The greengrocer takes a random sample of 10 oranges and records their weights in grams. The results are as follows.

176 194 193 191 170 177 178 160 188 169

(a) (i) Calculate unbiased estimates of the mean and variance of the weights of oranges, giving your answers to one decimal place.

(3)

(ii) Construct a 90% confidence interval for the mean weight of oranges.

(5)

(b) The greengrocer currently sells the oranges by weight at a price of £1.20 **per kilogram**.

(i) Deduce a 90% confidence interval for the mean price per orange.

(2)

(ii) The greengrocer decides to sell the oranges individually at a price of 20 pence per orange. Use the confidence interval you found in part (b)(i) to comment on this decision.

(2)

(Total 12 marks)

Q35.

An internet service provider operates a telephone helpdesk. The random variable T denotes the duration, in minutes, of a telephone call to the helpdesk. The internet service provider recorded the duration, t minutes, of each of a random sample of 50 telephone calls.

From the recorded durations of these telephone calls, the following values were calculated, where $\bar{t}$ denotes the sample mean.

$$\sum t = 143.50 \qquad \sum (t-\bar{t})^2 = 279.8929$$

(a) Calculate an unbiased estimate of Var(T) .and hence estimate the standard error of $\bar{T}$.

(3)

(b) (i) Construct a 99% confidence interval for the mean duration, μ minutes, of telephone calls to the helpdesk.

(5)

(ii) Hence comment on the claim that μ = 3.5.

(2)

(c) Three months later, from a random sample of 100 telephone calls, a 99% confidence interval for μ was determined correctly as (4.12, 5.38).

Indicate, giving reasons, what may now be concluded about the value of μ.

(2)

(Total 12 marks)

Q36.

Nuts are dispensed into tins by a machine that can be set to deliver any specified mean weight. For any setting on the machine, the weights of nuts per tin are normally distributed with a standard deviation of 3.0 grams.

Twelve tins of nuts are selected at random from a batch filled by the machine at a particular setting. The weights, in grams, are recorded as follows.

413.1	412.6	410.4	410.6	415.2	414.6
416.6	411.2	416.9	412.6	413.4	408.8

(a) Calculate, to one decimal place, the limits of a 99% confidence interval for the mean weight of nuts dispensed at this particular setting.

(5)

(b) The setting on the machine is marked as "410 grams". State, with a reason, whether this marking is likely to be correct.

(2)

(Total 7 marks)

Q37.

(a) Machine A produces plastic balls which have diameters that are normally distributed with a mean of 10.3 cm and a standard deviation of 0.16 cm.

(i) Determine the proportion of balls which have diameters less than 10.5 cm.

(3)

(ii) Calculate the diameter exceeded by 75 per cent of balls.

(4)

(b) Machine B produces beach balls which have diameters that are normally distributed with a mean of μ cm and a standard deviation of 0.24 cm.

The diameter, d cm, of each ball in a random sample of 144 beach balls was measured.

This gave:

$$\Sigma d = 3585.6.$$

(i) Calculate an unbiased estimate of μ.

(1)

(ii) Calculate the standard error of your unbiased estimate.

(2)

(iii) Construct a 95% confidence interval for μ, giving its limits to two decimal places.

(4)

(iv) Hence state, with a reason, whether you agree with a claim that $\mu = 25$.

(2)

(Total 16 marks)

Q38.

A medical statistician is studying the weights, x kg, of newborn babies in a hospital. She finds that, in a certain month, 25 babies were born in this hospital and that for these babies, $\Sigma x = 88.2$ and $\Sigma x^2 = 326.2$.

(a) (i) Calculate unbiased estimates of the mean and variance of the weights of babies born in this hospital.

(4)

(ii) Assuming that weights are normally distributed, calculate a 99% confidence interval for the mean weight of babies born in this hospital.

(4)

(b) She subsequently reads in a report that the mean weight of babies born in her region is 4.1 kg. Comment on her results with reference to this report.

(2)

(Ttoal 10 marks)

Q39.

An examination of the till roll at a petrol station showed that the previous 400 customers had spent a total of £4256 on petrol and that the standard deviation of the individual amounts spent on petrol was £3.68.

(a) Calculate a 95% confidence interval for the mean amount spent on petrol by a customer. Assume that the 400 customers may be regarded as a random sample of all customers using this petrol station.

(5)

(b) Further examination of the till roll showed that, although customers could spend any amount they chose on petrol, the great majority of customers spent a multiple of £5. It was suggested that an adequate model for the amount, in £, spent on petrol by customers could be provided by the discrete random variable, X, with the following probability distribution.

x	$P(X = x)$
5	0.15
10	0.63
15	0.15
20	0.07

Calculate:

(i) the mean of X;

(ii) $E(X^2)$.;

(iii) the standard deviation of X.

(6)

(c) Using the given data and the results of your calculations in parts (a) and (b), comment on the plausibility of X as a model for the amount spent by customers on petrol.

(3)

(d) Miguel, the petrol station owner, decided to offer a £1 token, redeemable on goods at the petrol station, to all customers who spent at least £12 on petrol. He hoped that most of his customers who would have spent £10 on petrol would then spend £15.

The first 200 customers, after the introduction of this offer, spent a total of £2342 on petrol and the standard deviation of the individual amounts spent on petrol was £3.42.

(i) Using the 5% significance level, examine whether the mean amount spent on petrol by a customer, following the introduction of this offer, exceeds £11. The 200 customers may be regarded as a random sample of all customers using this petrol station.

(8)

(ii) Explain to Miguel the implications of your result in part (d)(i).

(2)

(iii) What further information would you require before advising Miguel whether or not to continue with the offer?

(2)

(Total 26 marks)

Q40.

An examination of the till roll at a petrol station showed that the previous 400 customers had spent a total of £4256 on petrol and that the standard deviation of the individual amounts spent on petrol was £3.68.

(a) Calculate a 95% confidence interval for the mean amount spent on petrol by a customer. Assume that the 400 customers may be regarded as a random sample of all customers using this petrol station.

(5)

(b) In the hope of increasing his takings, Miguel, the petrol station owner, decided to offer a £1 token, redeemable on goods at the petrol station, to all customers who spent at least £12 on petrol.

The first 200 customers, after the introduction of this offer, spent a total of £2342 on petrol and the standard deviation of the individual amounts spent on petrol was £3.42.

(i) Calculate a 95% confidence interval for the mean amount spent on petrol by a customer following the introduction of the offer. Assume that the 200 customers may be regarded as a random sample of all customers using this petrol station.

(2)

(ii) Explain to Miguel the implications of your results in part (a) and part (b)(i).

(2)

(iii) What further information would you require before advising Miguel whether or not to continue with the offer?

(2)
(Total 11 marks)

Q41.

A company manufactures components for the motor industry. The components are designed to have a length of 135.0 mm. Each day a technician takes a random sample of components and calculates a 95% confidence interval for the mean length.

(a) On a particular Monday the technician takes a sample of nine components. Their lengths, in millimetres, are as follows.

135.1 135.7 134.9 135.2 136.3 135.7 135.9 136.0 135.6

Assuming the lengths of components produced on this Monday may be modelled by a normal distribution, calculate a 95% confidence interval for the mean length. Give your answer to an appropriate degree of accuracy.

(8)

(b) On a particular Friday a random sample of 60 components had a mean length of 135.8 mm and a standard deviation of 3.9 mm. Calculate a 95% confidence interval for the mean length of components produced on this Friday.

(4)

(c) The company overhauls all machines when the confidence interval calculated does not contain 135.0 mm. State whether increasing the sample size, n, is likely to increase, decrease or leave unaltered:

(i) the width of the confidence interval;

(1)

(ii) the frequency of machine overhauls, explaining your answer.

(2)

(d) Using your results in parts (a) and (b) as an example, or otherwise, explain why it is possible that production on a day when the confidence interval does not contain 135.0 mm will sometimes be more satisfactory than production on a day when the confidence interval does contain 135.0 mm.

(2)
(Total 17 marks)

Q42.

A company manufactures components for the motor industry. The components are designed to have a length of 135.0 mm. Each day a technician takes a random sample of components and calculates a 95% confidence interval for the mean length.

(a) On a particular Monday the technician takes a sample of nine components. Their lengths, in millimetres, are as follows.

135.1 135.7 134.9 135.2 136.3 135.7 135.9 136.0 135.6

Assuming the lengths of components produced on this Monday may be modelled by a normal distribution with standard deviation 0.42, calculate a 95% confidence interval for the mean length. Give your answer to an appropriate degree of accuracy.

(6)

(b) On a particular Friday a random sample of 60 components had a mean length of 135.8 mm and a standard deviation of 3.9 mm. Calculate a 95% confidence interval for the mean length of components produced on this Friday.

(2)

(c) The company overhauls all machines when the confidence interval calculated does not contain 135.0 mm. State whether increasing the sample size, n, is likely to increase, decrease or leave unaltered:

(i) the width of the confidence interval;

(1)

(ii) the frequency of machine overhauls, explaining your answer.

(2)

(d) Using your results in parts (a) and (b) as an example, or otherwise, explain why it is possible that production on a day when the confidence interval does not contain 135.0 mm will sometimes be more satisfactory than production on a day when the confidence interval does contain 135.0 mm.

(2)

(Total 13 marks)

Q43.

Applicants to join a police force are tested for physical fitness. Based on their performance, a physical fitness score is calculated for each applicant. Assume that the distribution of scores is normal.

(a) The scores for a random sample of ten applicants were

55 23 44 69 22 45 54 72 34 66

Calculate a 99% confidence interval for the mean score of all applicants.

(8)

(b) The scores of a further random sample of 110 applicants had a mean of 49.5 and a standard deviation of 16.5.

Use the data from this second sample to calculate:

(i) a 95% confidence interval for the mean score of all applicants;

(4)

(ii) an interval within which the score of approximately 95% of applicants will lie.

(2)

(c) By interpreting your results in parts (b)(i) and(b)(ii), comment on the ability of the applicants to achieve a score of 25.

(3)

(d) Give **two** reasons why a confidence interval based on a sample of size 110 would be preferable to one based on a sample of size 10.

(2)

(e) It is suggested that a much better estimate of the mean physical fitness of all recruits could be made by combining the two samples before calculating a confidence interval. Comment on this suggestion.

(3)

(Total 22 marks)

Q44.

Applicants to join a police force are tested for physical fitness. Based on their performance, a physical fitness score is calculated for each applicant. Assume that the distribution of scores is normal.

(a) The scores for a random sample of ten applicants were

55 23 44 69 22 45 54 72 34 66

Experience suggests that the standard deviation of scores is 14.8.

Calculate a 99% confidence interval for the mean score of all applicants.

(5)

(b) The scores of a further random sample of 110 applicants had a mean of 49.5 and a standard deviation of 16.5.

Use the data from this second sample to calculate:

(i) a 95% confidence interval for the mean score of all applicants;

(3)

(ii) an interval within which the score of approximately 95% of applicants will lie.

(2)

(c) By interpreting your results in parts (b)(i) and (b)(ii), comment on the ability of the applicants to achieve a score of 25.

(3)

(d) Give **two** reasons why a confidence interval based on a sample of size 110 would be preferable to one based on a sample of size 10.

(2)

(e) It is suggested that a much better estimate of the mean physical fitness of all recruits could be made by combining the two samples before calculating a confidence interval. Comment on this suggestion.

(3)

(Total 18 marks)

Q45.

A firm is considering providing an unlimited supply of free bottled water for employees to drink during working hours. To estimate how much bottled water is likely to be consumed, a pilot study is undertaken. On a particular day-shift, ten employees are provided with unlimited bottled water. The amount each one consumes is monitored. The amounts, in ml, consumed by these ten employees are as follows:

110	0	640	790	1120	0	0	2010	830	770

(a) Assuming the data may be regarded as a random sample from a normal distribution, calculate a 95% confidence interval for the mean amount consumed on a day-shift.

(8)

(b) (i) Give a reason, based on the data collected, why the normal distribution may not provide a suitable model for the amount of free bottled water which would be consumed by employees of the firm.

(ii) A normal distribution may provide an adequate model but cannot provide an exact model for the amount of bottled water consumed. Explain this statement giving a reason which does not depend on the data collected.

(3)

(c) Following the pilot study the firm offers free bottled water to all the 135 employees who work on the night-shift. The amounts they consume on the first night have a mean of 960 ml with a standard deviation of 240 ml.

(i) Assuming that these data may be regarded as a random sample, calculate a 90% confidence interval for the mean amount consumed on a night-shift.

(4)

(ii) Give **one** reason why it may be unrealistic to regard the data as a random sample of the amounts that would be consumed by all employees if the scheme was introduced on all shifts on a permanent basis.

(1)

(Total 16 marks)